

# Blake Grudzien

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## Experience

### Level Home

May 2025 – September 2025

*Software Engineer Intern | Case study*

*Redwood City, CA*

- Built the Go service behind a rebuilt device telemetry pipeline on AWS (Lambda, SQS, S3, Athena), cutting data loss from over 30% to under 1% across millions of events from smart locks and video doorbells
- Replaced silent deletion of unparseable files with a dead-letter queue, so the firmware team could ship a parser fix and clear the entire backlog with one redrive instead of a cross-team coordination cycle
- Chose SQS over EventBridge after finding fan-out irrelevant for a single-consumer path, and kept the native DLQ over a custom one, trading per-error retry control for zero custom code and one-command reprocessing
- Migrated telemetry storage from row-based JSON to Parquet via Apache Arrow so Athena scans only queried columns, choosing Arrow over DuckDB, whose CGo requirement would have forced changes across the service
- Built the service's test suite and CI pipeline, covering the full path from a raw telemetry file to queryable Parquet output
- Prototyped a Slack bot alerting on every parse failure, then narrowed it to a daily digest of the queue after the firmware team judged per-failure alerts to be noise

### Brooksee

June 2024 – October 2024

*Software Engineer Intern*

*Salt Lake City, UT (Hybrid)*

- Built a custom Garmin Connect IQ data field (MonkeyC) paired with a Kotlin Android app to stream live pace, distance, placement, and gun-time gaps to runners' watches during official Brooksee races (5,000–10,000 participants)
- Engineered a Firestore-to-Bluetooth telemetry pipeline with server-synced timing and fault-tolerant data parsing
- Derived each runner's elapsed time from the race's absolute start timestamp rather than a local timer, so runners crossing the line at irregular intervals after the gun still saw correct official time despite device clock drift

## Education

### Emory University

Atlanta, GA

*B.S. Computer Science, Minor in Mathematics | GPA: 3.65/4.00*

*May 2026*

- Coursework: Data Structures & Algorithms, ML, AI, Database Systems, Mathematical Statistics, Data Mining

## Technical Skills

**Languages:** Python, Go, Java, TypeScript, JavaScript, Kotlin, SQL, C++/C, R

**Frameworks & Libraries:** Flask, React, SvelteKit, Streamlit, Scikit-Learn, pandas, NumPy, pytest, D3.js

**Cloud & Data:** AWS (Lambda, SQS, S3, Athena, Kinesis), Parquet, PostgreSQL/pgvector, Firestore, Cloudflare Workers

**Tools & Other:** Git, GitHub Actions, Docker, Linux, REST APIs, RAG, vector embeddings, LLM tool-calling, Agile/Scrum

## Projects

### Chess Scholar ([Live](#) | [Github](#) | [Case study](#)) | *Python, Claude API, PostgreSQL/pgvector*

July 2026 – Present

- Built a hybrid retrieval system over 127,435 games and 10.2M moves (1610–2023), routing questions across deterministic SQL, pgvector search over 468,862 embedded annotation chunks, and pooled Stockfish subprocesses through Claude's native tool-calling rather than LangChain or a hand-rolled router
- Required engine grounding before any model claim about move quality, constraining the LLM to attribute answers to retrieved stats, prose, or evaluations rather than its own judgment
- Trained a gradient-boosted classifier (0.887 cross-validated ROC-AUC) to filter scraped study material before surfacing it, and covered the system with 235 tests running against Python 3.11 and 3.12 in CI
- Wrapped three paid APIs and Postgres in explicit error handling, with an 8 req/min session cap to stop runaway usage

### Bird Strike Analysis ([Live](#) | [Github](#)) | *SvelteKit, D3.js, TypeScript*

Feb 2026 – May 2026

- Built an interactive data-driven article analyzing 300k+ FAA bird strike records, featuring animated geographic maps, linked charts, and a species heatmap, deployed via GitHub Actions

### CHOA Chatbot ([Github](#)) | *Python, React, MongoDB*

Jan 2024 – May 2024

- Built an AI-powered chatbot for the Children's Hospital of Atlanta in a 7-person team, training resident doctors on pediatric airway management protocols for respiratory emergencies

## Leadership

### Vice President

Oct 2023 – Sep 2024

*Alpha Tau Omega Fraternity, Emory University (100+ members)*

*Atlanta, GA*

- Led chapter to 3rd highest GPA among all 136 national ATO chapters; served as head of disciplinary process, mediating conflicts across 100+ members

## Additional

**Chess:** 2209 chess.com elo (99.9 percentile) | **Languages:** Intermediate Spanish | **Interests:** Tennis, Rock Climbing, Soccer